Data Visualization

Lab 5 – Exploratory Data Analysis (EDA) Part (b) - Python

Dataset: “House Prices – Advanced Regression Techniques”

Source: Kaggle

Lab Type: Exploratory Data Analysis & Reporting

Duration: 3 Hours

Lab Purpose

- Handle missing values properly
- Detect outliers using IQR & Z-score
- Treat outliers
- Perform feature engineering
- Prepare cleaned dataset

Part 1 Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import skew, kurtosis
```

Explanation:

- NumPy → numerical operations
- Pandas → data handling
- Matplotlib/Seaborn → visualization
- SciPy → statistical metrics

Part 2 Loading Dataset

```
# Load dataset
df = pd.read_csv("train.csv")

# View first 5 rows
df.head()
```

Part 3 Handling Missing Values

3.1 Handling Missing Values

```
missing_percent = df.isnull().sum() / len(df) * 100
missing_percent.sort_values(ascending=False)
```

3.2 Visualize Missing Values

```
plt.figure(figsize=(12,6))
sns.heatmap(df.isnull(), cbar=False)
plt.show()
```

3.2 Imputation

a. Numerical Columns

```
num_cols = df.select_dtypes(include=['int64','float64']).columns

for col in num_cols:
    df[col].fillna(df[col].median(), inplace=True)
```

b. Categorical Columns

```
cat_cols = df.select_dtypes(include=['object']).columns

for col in cat_cols:
    df[col].fillna(df[col].mode()[0], inplace=True)
```

Part 4 Outlier Detection

4.1 IQR Method

```
Q1 = df[num_cols].quantile(0.25)
Q3 = df[num_cols].quantile(0.75)
IQR = Q3 - Q1

outliers = ((df[num_cols] < (Q1 - 1.5 * IQR)) |
            (df[num_cols] > (Q3 + 1.5 * IQR))).sum()
print(outliers.sort_values(ascending=False))
```

4.2 Z-Score Method

```
from scipy.stats import zscore

z_scores = np.abs(zscore(df[num_cols]))
outliers_z = (z_scores > 3).sum(axis=0)
print(outliers_z)
```

Part 5 Outlier Treatment

5.1 Winsorization

```
def cap_outliers(column):
    lower = column.quantile(0.01)
    upper = column.quantile(0.99)
    return np.clip(column, lower, upper)

for col in num_cols:
    df[col] = cap_outliers(df[col])
```

5.2 Log Transformation (for skewed features)

```
df['SalePrice'] = np.log1p(df['SalePrice'])
```

Part 6 Create Total Area Feature

6.1 Create Total Area Feature

```
df['TotalArea'] = df['TotalBsmtSF'] + df['1stFlrSF'] + df['2ndFlrSF']
```

6.2 Remove Highly Correlated Features

```
upper = corr_matrix.where(np.triu(np.ones(corr_matrix.shape), k=1).astype(bool))

to_drop = [column for column in upper.columns if any(upper[column] > 0.9)]

df.drop(columns=to_drop, inplace=True)
```

Part 7 Final Clean Dataset

```
df.to_csv("cleaned_house_prices.csv", index=False)
```

Part 8 Exercises

8.1 Exercise 1 - Compare Before & After Outlier Treatment

```
df.describe()
```

Write analysis.

8.2 Exercise 2 - Build Automated Outlier Function

```
def detect_and_treat_outliers(data):
# full automated function
return data
```

Student Instructions

1. Explain the significance of the insight in 3–5 sentences.
2. Insert the chart screenshot in the report.
3. Prepare a final report in MS Word or PDF format.
4. Submit ipynb file.